

1 Seed Environments

original seeds + retired environments

```
env_seed_math.py
_generate(...)
_prompt_generate(...)
scorer(...)
...
```

```
env_logic_grid.py
_generate(...)
_prompt_generate(...)
scorer(...)
...
```

```
env_sorting.py
_generate(...)
_prompt_generate(...)
scorer(...)
...
```

Joint RL Optimization (GRPO)

shared policy update



2 Shared Policy Model

same policy, two roles



Generator role

writes reusable executable environments

Solver role

solves prompts sampled from accepted environments

3 Candidate Environment

proposed by the generator

```
candidate_env.py
class SubsetSum_Environment(VerifiableEnvironment)
def _generate(seed, difficulty)
def _prompt_generate()
def _process(answer)
def scorer(output)
```

4 Validation & Admission

contract checks (admission gate)

Executable	L1-L5 validation
Semantically correct	semantic review
Frontier-calibrated	solver accuracy
Novel	novelty gate

Admitted env

5 Active Environment Pool

accepted reusable reward sources

Geometry TriangleCongruence_Environment.py CircleTangent_Environment.py ConvexHull_Environment.py ...
Number Theory ModularInverse_Environment.py GCDGame_Environment.py LinearDiophantine_Environment.py ...
Graph BridgeDetection_Environment.py ShortestPath_Environment.py BipartiteCheck_Environment.py ...
...

Sampled env

Compile to generate



```
$ python ./SubsetSum.py
```

Prompt 1: $S=\{3,4,8,10,13\}; T=21$. Return any subset summing to T .

Prompt 2: $S=\{6,9,11,15,20\}; T=26$. Return any subset summing to T .

Prompt n:

Solver Reward

verified solver performance



correctness / score

Environment Scorer

computes verified reward

Check: $8+13=21$ ✓ Reward: 1.0

Solver (shared policy)

generates answer

Answer 1: [8, 13]

retired environments reused as seeds (pool rotation)